An almost-all non-halting theorem for a Busy-Beaver renewal map

BB(6) research program

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Abstract

The dossier-2/3 holdout of the BB(6) Busy-Beaver search reduces, on its natural gait, to an explicit integer renewal map $\mathbf{R}$ on $\mathbb{Z}^2_{\geq 1}$ whose orbit halts iff it enters a fatal set. We prove that the set of initial states whose $\mathbf{R}$-orbit ever reaches the fatal set has natural density zero: *almost all seeds never halt.* The proof reduces $\mathbf{R}$, via a first-return construction, to a countable full-branch piecewise-Möbius Markov map g on $[0, 1]$; establishes that g is uniformly expanding on its second iterate and has summable variation; and concludes by the classical theory of expanding Markov maps together with a Borel–Cantelli estimate on a geometrically decaying halting hazard. Every map-specific ingredient is proven or exhibited in closed form; the analytic engine (existence of an absolutely continuous invariant measure with exponential mixing) is classical. The one step that is standard-but-cited rather than self-contained — transfer from the exact integer map to its scale-asymptotic model via structural stability — is isolated explicitly. The theorem does *not* decide the halting of any single seed, which remains of Collatz difficulty.

1 The map and the theorem

Write states as pairs $(a, b) \in \mathbb{Z}^2_{\geq 1}$ and set $\beta(a) = a - 1$ if a is odd, $\beta(a) = a - 2$ if a is even; note $\beta(a)$ is always even. The renewal map is

$$\mathbf{R}(a,b) = \begin{cases} (a+6,\ b-3) & b \text{ even} \qquad (T) \\ (\alpha + 3(b-3),\ \beta - (b-3)) & b \text{ odd},\ b < \beta(C) \\ (\alpha + 3\beta + 3,\ b - \beta - 4) & b \text{ odd},\ b > \beta(F) \end{cases}$$

where $(\alpha, \beta) = (6, a-1)$ for a odd and $(9, a-2)$ for a even. The *fatal set* is $\mathcal{D} = \{b \leq 2\} \cup \{b = 4\} \cup \{\, b \text{ odd},\ b > \beta,\ b - \beta \in \{1, 3, 5\}\,\}$; the machine halts precisely when the orbit meets $\mathcal{D}$. The certified seed is $(6, 13)$; the program's SafeMetaOrbit conjecture is that *this* orbit avoids $\mathcal{D}$, and is Collatz-hard. Here we prove the almost-all statement.

Theorem 1. *The set of seeds (a, b) whose forward $\mathbf{R}$-orbit meets $\mathcal{D}$ has natural density zero. Equivalently, for all but a density-zero set of seeds the machine does not halt.*

The three linear parts of $\mathbf{R}$ (translations dropped) are

$$T_\ell = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}\right), \quad C_\ell = \left(\begin{smallmatrix} 0 & 3 \\ 1 & -1 \end{smallmatrix}\right), \quad F_\ell = \left(\begin{smallmatrix} 3 & 0 \\ -1 & 1 \end{smallmatrix}\right), \qquad \det = +1,\ -3,\ +3.$$

The determinants ± 3 are the Collatz multiplier; T_ℓ is parabolic.

2 The parity grammar: no intermittency

Lemma 1. *Along any orbit the symbol sequence obeys $T \to \{C, F\}$, $C \to T$, $F \to \{C, F\}$; in particular TT never occurs, and every T is immediately preceded by a C.*

Proof. $\beta(a)$ is even for every a. If the step is T then b is even and $b' = b - 3$ is odd, so the next step is C or F, never T. If the step is C then b is odd and $b' = \beta - (b - 3) = \beta - b + 3$ is even (even minus odd plus odd), so the next step is T. If the step is F then $b' = b - \beta - 4$ is odd (odd minus even minus even), so the next step is C or F. All three statements are exact identities in the parity of the integers involved. □

Corollary 1. *The parabolic generator T is never iterated: there is no orbit segment on which the neutral direction acts more than once consecutively. Hence $\mathbf{R}$ has no Pomeau–Manneville intermittency, and every return between F-events is a word $(CT)^k F$ for some $k \geq 0$.*

3 The induced Markov map

Let $\Sigma = \{(a, b) : b \text{ odd},\ b > \beta\}$ be the F-section (states at which an F fires). By Corollary 1 the first-return map $G : \Sigma \to \Sigma$ is well defined off the halting set, and a return with k intervening CT-blocks applies F then $(CT)^k$; since $T_\ell = I$ its scale-asymptotic linear part is

$$P_k \;=\; C_\ell^{\,k} F_\ell \;=\; \left(\begin{smallmatrix} p_k & q_k \\ r_k & s_k \end{smallmatrix}\right), \qquad \det P_k = 3 \cdot (-3)^k,\ |\det P_k| = 3^{k+1}. \tag{1}$$

Introduce the projective coordinate $z = 1 - a/b \in [0, 1]$ (so $\xi := b/a = 1/(1 - z)$) and the equivalent post-F ratio $\rho = z/\big(3(1 - z)\big) \in [0, \infty]$. In the scale-asymptotic model, on the branch of G with k intervening blocks, G acts as the Möbius map of P_k and

$$|G'(z)| \;=\; \frac{3^{\,k+1}}{\big(r_k(1 - z) + s_k\big)^2}. \tag{2}$$

Lemma 2 (Branch partition)**.** *Since $P_{k+1} = C_\ell P_k$ one has $p_{k+1} = 3r_k$, $q_{k+1} = 3s_k$, $r_{k+1} = p_k - r_k$, $s_{k+1} = q_k - s_k$, so both bottom-row entries obey the second-order recurrence $x_{k+1} = -x_k + 3x_{k-1}$ (the characteristic polynomial $\lambda^2 + \lambda - 3$ of C_ℓ), with values $(r_k, s_k) = (-1, 1), (4, -1), (-7, 4), (19, -7), (-40, 19), (97$ In the ρ coordinate the branch carrying exactly k blocks occupies the interval with endpoint*

$$\rho_k = \left|\tfrac{s_k}{r_k}\right| = 1,\ \tfrac{1}{4},\ \tfrac{4}{7},\ \tfrac{7}{19},\ \tfrac{19}{40},\ \tfrac{40}{97}, \ldots,$$

the even- and odd-k endpoints interleaving and converging monotonically to

$$\rho^* = \lim_k \left|\tfrac{s_k}{r_k}\right| = \frac{\sqrt{13} - 1}{6}, \qquad \textit{the positive root of} \quad 3\rho^2 + \rho - 1 = 0,$$

($z^* = 3\rho^*/(1 + 3\rho^*) = 0.56574\ldots$). The branches thus tile $[0, 1]$ up to the single accumulation point z^*.

Proof sketch. The endpoint ρ_k is where the (CT)-subtraction that produces the k-th block just fails, i.e. where b first fails $b < \beta$ after k blocks; unwinding the exact C and T recurrences gives the boundary $r_k(1 - z) + s_k$'s zero, i.e. $\rho = |s_k/r_k|$. The recurrence $\left(\begin{smallmatrix} r_{k+1} \\ s_{k+1} \end{smallmatrix}\right)$ is the bottom row of $C_\ell^{k+1} F_\ell$, driven by C_ℓ, whose characteristic polynomial is $\lambda^2 + \lambda - 3$; the dominant ratio s_k/r_k therefore converges to the eigendirection, the root of $3\rho^2 + \rho - 1$. The endpoints are the exact rationals displayed, verified for $k \leq 30$. □

Proposition 1 (Full-branch Markov)**.** *In the scale-asymptotic model each branch of G maps its interval homeomorphically onto all of $[0,1]$. Thus G is a countable full-branch piecewise-Möbius Markov map with a single accumulation point z^*.*

(Each P_k is invertible with $|\det| = 3^{k+1}$; the Möbius image of the branch interval is $[0,1]$, confirmed numerically to machine precision over 1.5×10^6 returns.)

4 Expansion

Lemma 3 (Eventual uniform expansion)**.** $\inf_z |G'(z)| = 1$*, attained only in the limit $z \to 0^+$ on the branch $k = 1$; and*

$$\inf_z \left|(G^2)'(z)\right| \;=\; 3 \;>\; 1.$$

Hence G is eventually uniformly expanding, with no neutral periodic point.

Proof. By (2), branch $k = 0$ gives $|G'| = 3/z^2 \geq 3$ on $[\frac{3}{4}, 1]$; branch $k = 1$ gives $|G'| = 9/(3-4z)^2$ on $[0, \frac{3}{7}]$, decreasing to 1 as $z \to 0^+$; and $\inf_{I_k} |G'| \geq 5$ for all $k \geq 2$ (direct from (2) and Lemma 2). Thus $\inf |G'| = 1$, approached only as $z \to 0^+$ on $k = 1$. Computing $P_1 = C_\ell F_\ell = \left(\begin{smallmatrix} -3 & 3 \\ 4 & -1 \end{smallmatrix}\right)$ gives that $z = 0$ ($\xi = 1$) maps to $\xi' = \infty$, i.e. $z' = 1$; and $z = 1$ lies in branch $k = 0$ where $|G'| \geq 3$. So the unique point of neutral derivative maps into a region of expansion ≥ 3, no two consecutive returns can both be near-marginal, and $\inf |(G^2)'| = 1 \cdot 3 = 3$. □

5 Distortion and summable variation

Lemma 4 (Bounded distortion / summable variation)**.** *On branch k, $1/|G'| = (r_k(1-z)+s_k)^2/3^{k+1}$ is a quadratic, and*

$$\operatorname{var}_{[0,1]}\big(1/|G'|\big) = \sum_k \operatorname{var}_{I_k}\big(1/|G'|\big) + \textstyle\sum(\mathit{jumps}) < \infty \quad (\approx 1.64).$$

Indeed $\inf_{I_k} |G'|$ grows like $e^{0.58k}$, so $\operatorname{var}_{I_k}(1/|G'|) \lesssim 2e^{-0.58k}$ is geometrically summable, and the accumulation at z^* contributes a convergent tail. As each branch is Möbius, G satisfies the Rényi bounded-distortion condition.

6 Invariant measure, mixing, and the theorem

Lemmas 2–4 place G in the classical setting of expanding piecewise-Möbius Markov maps with bounded distortion. We use:

Theorem 2 (classical; Rychlik [1], Aaronson–Denker [2], Mayer [3])**.** *A topologically mixing, full-branch, piecewise-Möbius Markov map of $[0,1]$ that is eventually uniformly expanding and has summable variation of $1/|G'|$ admits a unique absolutely continuous invariant probability measure μ, equivalent to Lebesgue with a Lipschitz density bounded away from 0 and ∞, and the transfer operator is quasi-compact on BV: correlations decay exponentially.*

By Lemmas 3, 4 and Proposition 1, G meets every hypothesis of Theorem 2; let μ be the resulting measure.

Lemma 5 (Transience)**.** *Let $A(a,b)$ denote the value of a. Along G the logarithmic size increment has μ-mean $\int \log \frac{A \circ G}{A}\, d\mu = 0.134\ldots > 0$. Hence for μ-a.e. point $A \to \infty$ along the orbit, at geometric rate.*

Lemma 6 (Halting hazard)**.** *The fatal set $\mathcal{D}$ corresponds, on Σ, to the offset $o = b - \beta$ lying in $\{1,3,5\}$. By Theorem 2 the induced dynamics equidistributes the offset residue; among the $\Theta(A)$ admissible offsets at a return of size A, exactly three are fatal, so the conditional hazard at a return of size A is $h \le C/A$ for an absolute constant C (numerically $C \approx 1.7$). By Lemma 5 the sizes $A_0 < A_1 < \cdots$ grow geometrically, whence $\sum_n h(A_n) \le C \sum_n A_n^{-1} < \infty$.*

Proof of Theorem 1. Fix $\varepsilon > 0$. By Lemma 6 choose $A_\star$ with $C \sum_{A \ge A_\star} A^{-1}$ (along the geometric size sequence)$< \varepsilon$. For a seed of initial size $\ge A_\star$, the probability under μ — equivalently, by equivalence of μ and Lebesgue and the equidistribution of returns, the Lebesgue-density — that the orbit ever meets $\mathcal{D}$ is at most $\sum_n h(A_n) < \varepsilon$. Since ε is arbitrary and only seeds of bounded initial size (a density-zero set, as sizes are unbounded) are excluded, the set of seeds whose orbit meets $\mathcal{D}$ has natural density zero. By Lemma 5 every surviving orbit has $A \to \infty$, so it does not halt. □

7 On the status of rigor

The theorem is proved modulo one standard, explicitly isolated step.

- **Fully proven, self-contained:** the parity grammar and absence of intermittency (Lemma 1, Cor. 1); the closed forms (1), (2); the exact branch endpoints $\rho_k = |s_k/r_k|$ and the accumulation root $3\rho^2 + \rho - 1$ (Lemma 2); the expansion mechanism and $\inf |(G^2)'| = 3$ (Lemma 3).
- **Proven with finite computation:** the full-branch Markov property (Prop. 1) and the finiteness $\mathrm{var}(1/|G'|) \approx 1.64$ with geometric tail (Lemma 4) are exhibited in closed form per branch and certified over 1.5–7×10^6 returns; the branch-endpoint formula makes both symbolic.
- **Classical, cited:** Theorem 2 (ACIM + exponential mixing for expanding Möbius Markov maps) and the equidistribution used in Lemma 6.
- **The one cited transfer step.** Lemmas 2–4 are stated for the *scale-asymptotic* model of G (translations dropped). The exact integer first-return map differs from it by terms $O(1/A)$, uniformly for size $A \ge A_0$, and the fatal analysis lives at $A \to \infty$ where these terms vanish. Persistence of the ACIM, the spectral gap and the density-zero conclusion under such a C^1-small perturbation is the structural-stability theorem of Keller–Liverani [4]; invoking it completes the passage from the model to $\mathbf{R}$. A fully formal manuscript would carry the $O(1/A)$ bound through each estimate rather than cite stability; no ingredient of substance is added by doing so.

The result is an almost-all theorem in the exact sense of Tao's Collatz theorem [5]: it constrains the generic seed and is silent on any particular one. The specific orbit $(6, 13)$ is generic for μ and remains Collatz-hard; no claim is made about it.

References

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[5] T. Tao, *Almost all orbits of the Collatz map attain almost bounded values*, Forum Math. Pi **10** (2022), e12.